Directed Studies Course Proposal (CS6999)

Faculty of Computer Science

Title: Robust Machine Learning
Instructor: Dr. Roozbeh Razavi-Far
Semester: Summer 2026
Students: **Refat Ishrak Hemel**

Course Description: This course will first focus on the topic of robustness of machine learning models, and then, study the fairness of machine learning algorithms. We will discuss the fundamentals of each topic as well as corresponding open challenges, with a focus on the algorithms. Students will understand different machine learning algorithms, analyze their implementation, robustness, and fairness through a series of readings and projects, and develop the ability to conduct research projects on related topics.

Course Content: Background ideas about robust machine learning including:

Robust Machine Learning:
Robust optimization and regularization,
Robustness of graph neural networks,
Robustness in distributed learning,
Techniques for robust model evaluation,
Robust data valuation,
Robust reinforcement learning,
Fairness and privacy of machine learning models:
Differential privacy in machine learning models,
Differential privacy in graphs,
Differential privacy in reinforcement learning,
Fairness in machine learning.

Course Objectives: By the end of this course, students will be able to understand the principles of robust machine learning. At the end of this course, students will be able to: (1) analyze machine learning models' robustness and develop algorithms to enhance their robustness; and (2) analyze, design, and develop algorithms to ensure fairness in machine learning.

Grading:

Weekly Oral Discussions/Presentations (30%)
Literature Survey and Project Report (30%)
Project Implementation (30%)
Final Presentation (10%)

Weekly presentation and discussion: Students will read several relevant papers over the semester. They will learn about most recent advancements on the proposed topics. During the course, students are expected to have weekly meetings with the instructor to discuss their progress. In these meetings the student will discuss/present about the paper they read or the framework they worked on.

Literature survey and report: Students will write a literature survey and report on a selected topic relevant to the course. This would summarize the existing research on that topic and suggest possible future works.

Project: The project will be chosen by the student a relevant topic. The student will be expected to contribute by implementing a software system and conduct an evaluation. The grade for this aspect of the project will be based on the software implementation, evaluation results and significance of the project.

Final Presentation: The final oral presentation will present the findings of the background research as well as the results of the project.

References:

[1] J. Lu, T. Issaranon, and D. Forsyth, "Safetynet: Detecting and rejecting adversarial examples robustly," 2017.

[2] A. Raghunathan, J. Steinhardt, and P. Liang, "Certified defenses against adversarial examples," in International Conference on Learning Representations, 2018.

[3] C. Liu, B. Li, Y. Vorobeychik, and A. Oprea, "Robust high-dimensional linear regression," 2016.

[4] D. Zhu, Z. Zhang, P. Cui, and W. Zhu, "Robust graph convolutional networks against adversarial attacks," in Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD 2019, Anchorage, AK, USA, August 4-8, 2019, pp. 1399–1407.

[5] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," 2017.

[6] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security.

[7] H. H. Nguyen, A. Imine, and M. Rusinowitch, "Detecting communities under differential privacy," 2016.
[8] P. Gajane and M. Pechenizkiy, "On formalizing fairness in prediction with machine learning," 2018.

[9] Ilias Diakonikolas, and Daniel M. Kane. "Recent advances in algorithmic high-dimensional robust statistics." arXiv preprint arXiv:1911.05911, 2019.

[10] Jacob Steinhardt, Pang Wei W. Koh, and Percy S. Liang. "Certified defenses for data poisoning attacks." In Advances in neural information processing systems, 2017.

[11] Nicholas Carlini, and David Wagner. "Towards evaluating the robustness of neural networks." In IEEE symposium on security and privacy (SP), 2017.

[12]Zico Kolter and Aleksander Madry, "Adversarial Robustness - Theory and Practice." 2018.

[13] Alexandra Chouldechova, and Aaron Roth. "The frontiers of fairness in machine learning." arXiv preprint arXiv:1810.08810, 2018.

[14] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. "Fairness through awareness." In Proceedings of the 3rd innovations in theoretical computer science conference, 2012.

[15] Reza Shokri, Martin Strobel, and Yair Zick. "On the Privacy Risks of Model Explanations." AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2021.

[16] Neel Patel, Reza Shokri, and Yair Zick. "Model Explanations with Differential Privacy." arXiv preprint arXiv:2006.09129, 2020.

Note: This reference list is not complete, but includes a selection of books, papers, and journal articles that are applicable to the course topics. The students will be responsible for performing the literature search on the area of Trustworthy ML that is of interest to them from various key conferences and journals.